Hallucination Failure Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Hallucination Failure Types before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Hallucination Failure Types trace and record the first step you can explain without help.

Use W2: Trace the worked case for Generation failures and verification. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Hallucination Failure Types explanation against the supplied facts before discussing it.

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Hallucination Failure Types target to a trace and a justified conclusion.

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Hallucination Failure Types case and show the deciding evidence.

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 14 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Hallucination Failure Types adds to Verified Study Buddy and keep the evidence.

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Hallucination Failure Types facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Hallucination Failure Types", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Hallucination Failure Types facts and keep your

> reasoning.

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Hallucination Failure Types facts and keep your reasoning.

A report omits this qualification: Source agreement and arithmetic validity are separate checks; missing **evidence** is not proof of the opposite claim.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Hallucination Failure Types facts and keep your reasoning.

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Hallucination Failure Types facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Hallucination Failure Types", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Hallucination Failure Types

> A fictional Study Buddy drafts a field-trip summary from one approved record. Explain the labelled step before checking the changed condition.

A fictional Study Buddy drafts a field-trip summary from one approved record.

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

AI ACADEMY / AI-3/14

Hallucination Failure Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Compute duration

12 / 4=3 hours, excluding stops.

↓

2. Contradicted claim

Two-hour duration disagrees with the calculation from S.

↓

3. Supported claim

Departure 09:00 is stated in S.

↓

4. Unsupported claim

Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

CHANGE ONE THING

Add a planned 30-minute stop while preserving speed and distance.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can we conclude there will definitely be no lunch from S alone?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy drafts a field-trip summary from one approved record.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

1. **Unsupported claim**

2. Supported claim

3. **Contradicted** claim

4. Compute duration

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Unsupported claim** Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Different errors need different repairs.

Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a planned 30-minute stop while preserving speed and distance. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Planned total elapsed time becomes 3.5 hours.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Repair a missing building block

Review Multimodal **Evidence** (ai-3/13). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Why were both “exactly three” and “exactly five” **unsupported** before opening the box?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

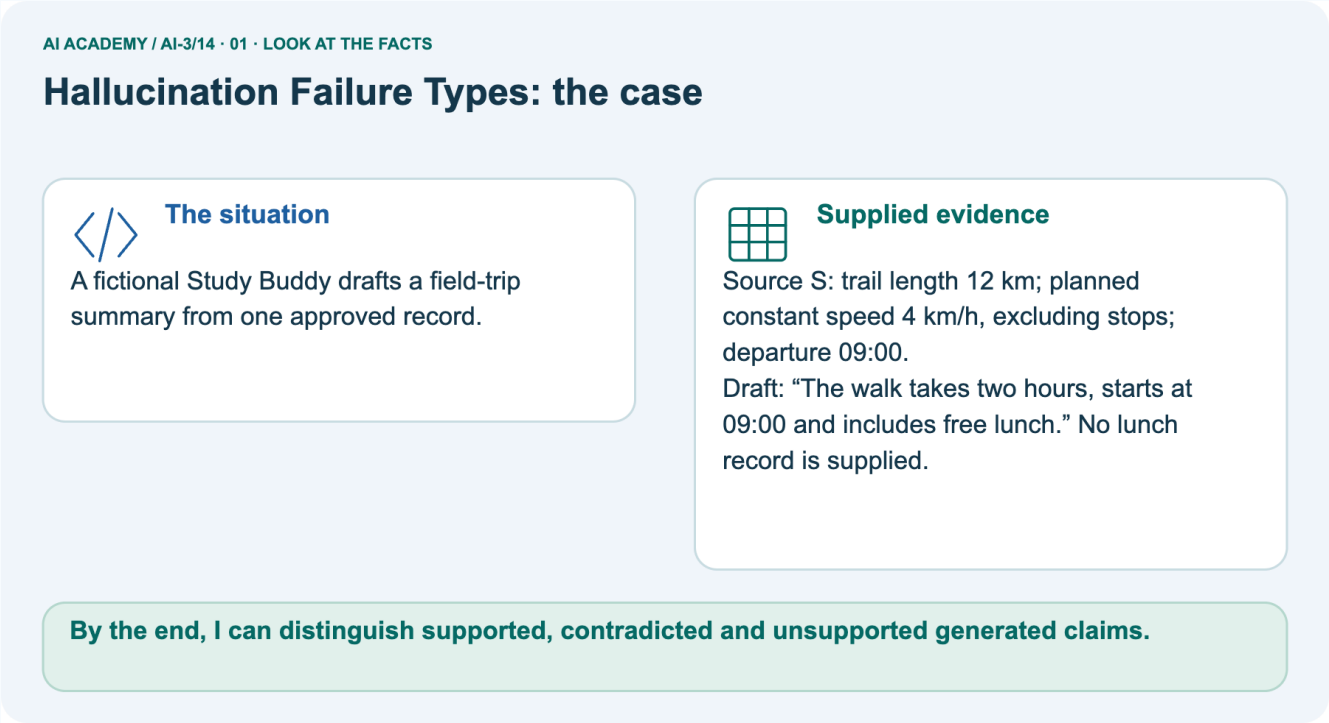
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Hallucination Failure Types picture from its labels, then record a relevant limit.

01. Hallucination Failure Types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

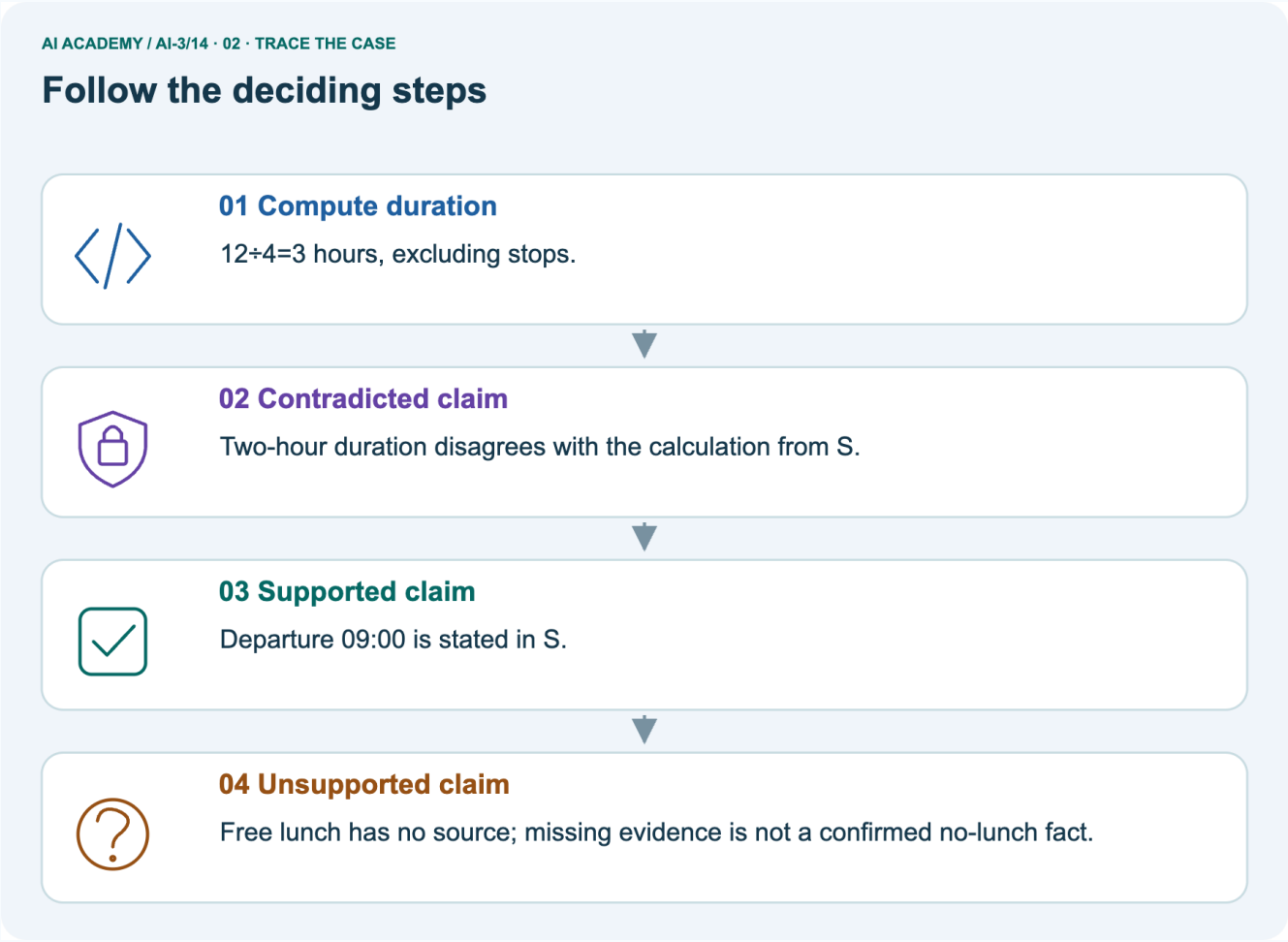
A fictional Study Buddy drafts a field-trip summary from one approved record. Supplied facts: Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



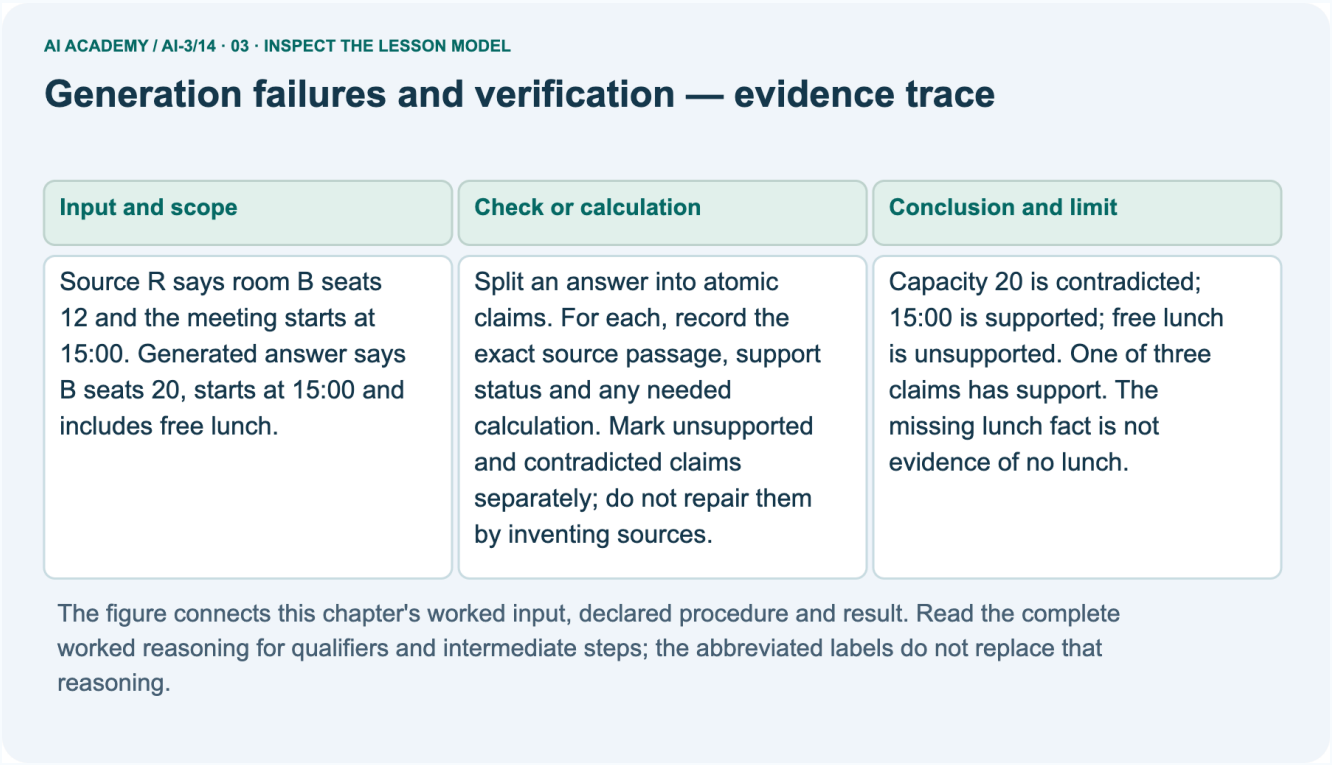
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute duration: 12÷4=3 hours, excluding stops. → **Contradicted** claim: Two-hour duration disagrees with the calculation from S. → Supported claim: Departure 09:00 is stated in S. → **Unsupported claim**: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Generation failures and verification — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generation failures and verification — **evidence** trace. Source R says room B seats 12 and the meeting starts at 15:00.

Generated answer says B seats 20, starts at 15:00 and includes free lunch. | Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation.

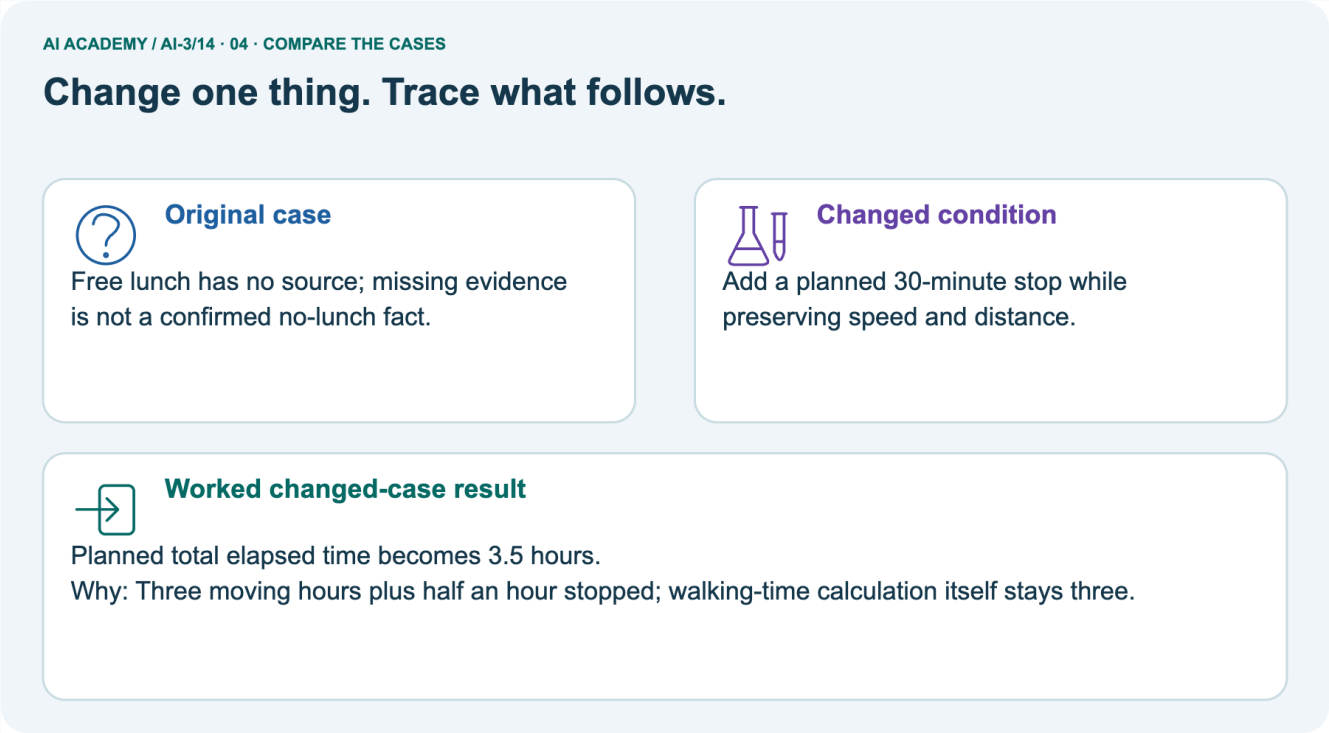
Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources. | Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support.

The missing lunch fact is not **evidence** of no lunch.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

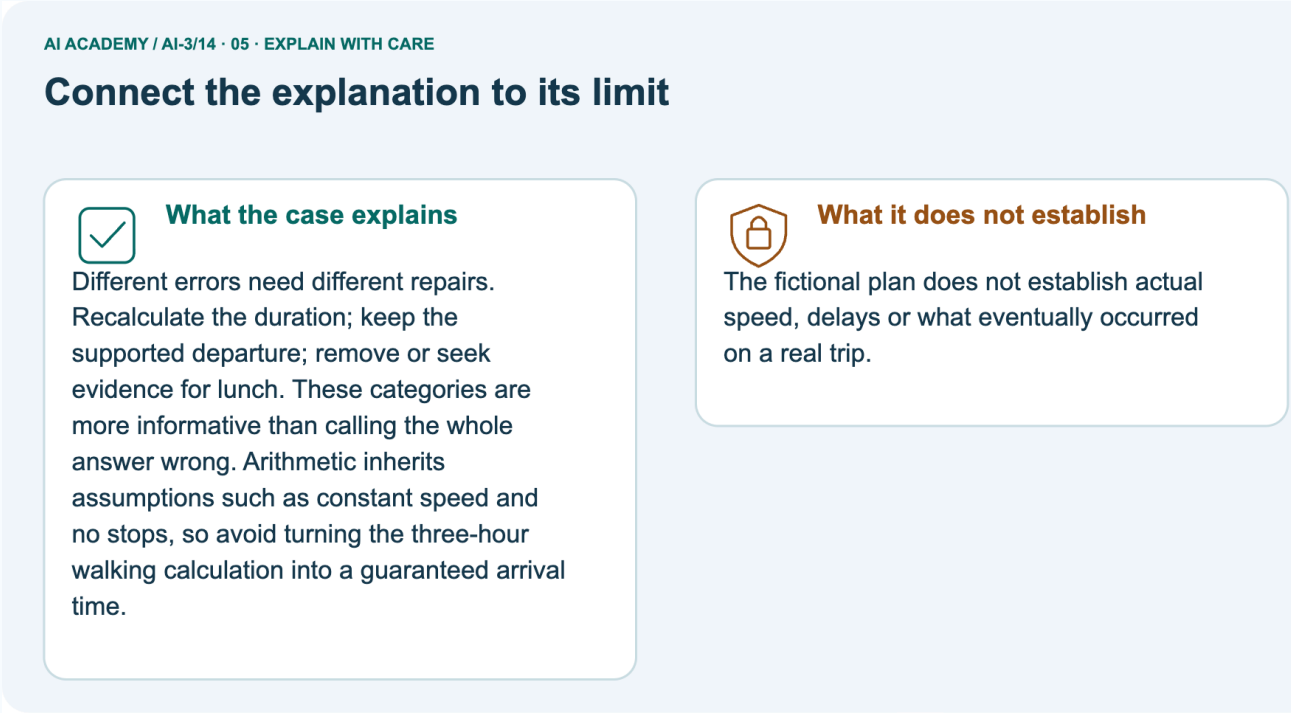
Original case: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Changed condition: Add a planned 30-minute stop while preserving speed and distance.

Expected result: Planned total elapsed time becomes 3.5 hours. Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

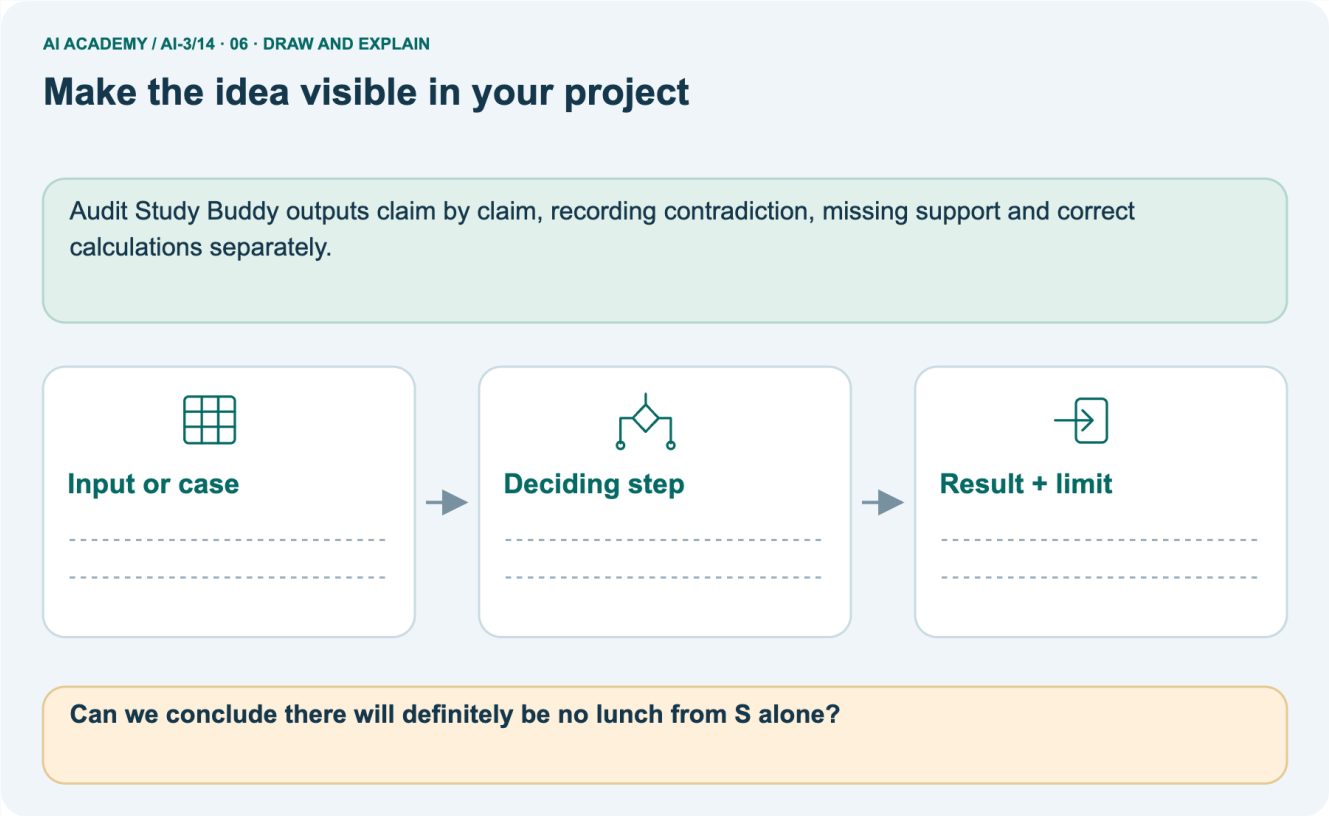
Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Limit: The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately. Draw your **input**, deciding step, result and limit.

Supported question: Can we conclude there will definitely be no lunch from S alone?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.